

XRPL Validator — Daily Engineering Report • 2026-09-25

Generated 2026-09-26 00:28 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **6 commits** (xrpl-validator 6)
- **2 engine findings fixed** (F401–F402)
- 1 test / vector commit
- Live shadow: 11,965/11,965 ledgers fully matched, 1,573,232 transactions, 0 engine receipts in the current soak
- Amendment **fixBatchV1_2** gained majority at ledger 107,227,649 (2026-09-25 14:13 UTC)
- Amendment **BatchV1_1** lost majority at ledger 107,227,905 (2026-09-25 14:29 UTC)
- Amendment **BatchV1_1** gained majority at ledger 107,228,161 (2026-09-25 14:46 UTC)
- Amendment **PermissionDelegationV1_1** held majority (since 2026-09-24 21:25 UTC); activation due ~2026-10-08 21:25 UTC
- Amendment **fixBatchV1_2** held majority (since 2026-09-25 14:12 UTC); activation due ~2026-10-09 14:12 UTC
- Amendment **BatchV1_1** held majority (since 2026-09-25 14:46 UTC); activation due ~2026-10-09 14:46 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	11,965
Ledgers fully matched	11,965
Transactions applied natively	1,573,232
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-09-25 11:33 EDT
Reference rippled	3.4.1, full, validated #107,240,858

Engine findings fixed

- **fix(ledger): F401 — AMM withdrawal reserve check counts XRP already paid out (soak #29 receipt)**
(166f033 , xrpl-validator) rippled's AMMWithdraw::withdraw checks the reserve for a new trust line against

max(priorBalance, balance) (AMMWithdraw.cpp:705), where balance is the account's live balance. It pays the transaction's Asset before it checks Asset2, so when Asset is XRP the check sees the XRP already withdrawn. withdraw_reserve_ok compared the pre-fee balance alone, which is exact for a single-asset withdrawal but not for a two-asset ...

- **fix(ledger): F402 — TicketCreate paid with a Ticket does not count the ticket it spends** (e63a7d9 , xrpl-validator) rippled's Transactor::apply runs consumeSeqProxy before doApply: a transaction paid for with a Ticket deletes that ticket first, lowering OwnerCount and TicketCount by one. TicketCreate's checks see the account after that: doApply requires the pre-fee balance to cover accountReserve(OwnerCount + TicketCount), and preclaim refuses with tecDIR_FULL only when curTicketCount + addedTickets - consumedTickets exceeds 250. ...

Engineering changes

- **fix(node): validation relay answers the hubs' keep-alive pings** (458c2c7 , xrpl-validator) live_viewer's validation relay (val-sender) read and discarded everything the relay hubs sent, including their keep-alive pings. rippled pings each peer on a 60-second timer and drops one that has not answered the previous ping by the next tick (PeerImp::onTimer, "Ping Timeout"), so every relay connection closed about two minutes after it opened, and the relay learned of it only when its next write failed. On ...
- **fix(node): voting-ledger validations keep their fields in canonical order** (1c373c9 , xrpl-validator) rippled deserializes a peer's validation with requireCanonicalOrder (PeerImp::onMessage(TMValidation) -> STObject::set; present in 3.3.0 and 3.4.0, absent in 3.1.2): each field's code must exceed the previous one, or the validation is dropped ("Fields in object are not in canonical order") and the sender is charged for invalid data. sign_validation appended the Signature (VL, field 6) after every other field. On the ...
- **feat(node): stop voting for SingleAssetVault and LendingProtocol** (0c76af1 , xrpl-validator) Either amendment switches rippled's Number to its 19-digit mantissa for every transaction: setCurrentTransactionRules (Rules.cpp) selects the Large mantissa scale when SingleAssetVault or LendingProtocol is enabled. The engine models the 16-digit scale only, so this validator no longer votes for either. Decision of the validator's operator, 2026-09-25.

Tests and campaigns

- **test(node): campaign 25 — Batch x Clawback, AMMClawback and token escrow (devnet vectors)** (677c18d , xrpl-validator)

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Deployment and the first warm restart.** Build 166f033 (findings F401, the voting-ledger validation order and the validation relay, all described below) replaced the live build at 09:46 EDT after its gate (1,636 suite tests and 707 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches). It was the first deployment to use the warm restart merged on 2026-09-24: the stopping build recorded its last verified ledger, and the new build rebuilt its stored state in 22.1 seconds, confirmed that the state's root equals the network's account hash for ledger 107,227,250, and resumed two ledgers behind the network. Cold starts had

resumed 90–111 ledgers behind, several minutes after the switch. The health check matched 400 ledgers with no mismatches; soak #30 began at 10:14 EDT.

- **Voting-ledger validations.** Every 256th ledger (sequence $\equiv 255 \bmod 256$) is a voting ledger: a validator's validation of it carries its amendment votes. The node signed those validations but placed the signature after the amendment list, which puts the fields out of canonical order, and rippled drops a validation whose fields are out of order (the check is present in rippled 3.3.0 and 3.4.0). The reference node's record from 2026-09-24 12:48 to 2026-09-25 00:22 EDT has our validation for none of the 41 voting ledgers in that period, so the network never received our amendment votes; it has our validation for every other ledger after 21:00 on 2026-09-24. The signature now takes its canonical place; the signed content is unchanged. Before deployment this was verified on mainnet with a throwaway key: the same voting-ledger validation, sent through a public hub, reached the reference node 93 ms later in the corrected layout and did not arrive in the old one. After deployment the reference node received our validation of voting ledger 107,227,391 at 09:56 EDT, the first voting-ledger validation of ours in its record. Fixed in 1c373c9.
- **Amendment votes.** The validator no longer votes for SingleAssetVault or LendingProtocol. Either amendment switches rippled's `Number` arithmetic to a 19-digit mantissa for every transaction, and the engine models the 16-digit scale that mainnet uses. Deployed at 10:53 EDT (below); from voting ledger 107,228,415 the validator's votes cover 46 amendments instead of 48.
- **Validation relay.** Besides its regular peer connections, the node sends its validations over four dedicated connections to well-connected hub servers. Those connections read and discarded everything the hubs sent, including their keep-alive pings. rippled pings each peer on a 60-second timer and drops a peer that has not answered its previous ping by the next tick, so each relay connection closed about two minutes after it opened; on 2026-09-24 the four relays reconnected 87 times in one hour. These drops did not cost validations at the reference node, which received every non-voting validation after 21:00 through the other connections. The relays now answer the pings, notice a closed connection at once, and after reconnecting re-send the last validation if it went out less than 60 seconds earlier. Fixed in 458c2c7 and deployed at 09:46; from the warm restart at 11:33 to the end of the day, each of the three hubs that accept peer connections held a single relay session with no drop. The fourth hub no longer accepts peer connections and will be replaced.
- **Soak #29 receipt and finding F401.** At ledger 107,219,029 an AMMWithdraw (withdraw all, from an XRP/DROP pool) came from an account holding 2.310461 XRP at an owner count of 6 with no DROP trust line; a new line at that owner count needs a 2.4 XRP reserve. rippled checks that reserve against the larger of the balance before the fee and the account's current balance, and it pays the XRP side out before it checks the DROP side, so the check saw 4.753549 XRP and mainnet accepted the withdrawal. The check added for F399 on 2026-09-24 compared only the balance before the fee and refused it. Fixed in 166f033 (deployed above).
- **Finding F402.** At ledger 107,227,378, during the new build's health check, a TicketCreate of 10 tickets paid for with a Ticket came from an account holding 3.398367 XRP at an owner count of 2. rippled deletes the ticket being spent before TicketCreate checks its reserve, so ten more tickets needed $1 + 0.2 \times 11 = 3.2$ XRP and mainnet succeeded; the engine counted the spent ticket, required 3.4 XRP and refused. The same account repeated the pattern at ledger 107,227,684. Fixed in e63a7d9, with the mainnet transaction pinned as a regression vector, and deployed at 10:53: the same account's next ticket-funded TicketCreate inside the affected balance range, at ledger 107,228,620, matched mainnet.
- **Campaign 25 (devnet).** Forty byte-exact vectors for Batch (BatchV1_1), combined with trust-line and MPT clawback, AMMClawback, and MPT and IOU escrow, all already enabled on mainnet. Thirty-seven are byte-exact under both flow engines, with no engine finding. The other three differ only because devnet enables

SingleAssetVault and LendingProtocol and so runs `Number` at 19 digits; libxrpl 3.4.0 under devnet's rules reproduces devnet's results. Merged as 677c18d.

- **fixBatchV1_2.** An amendment named fixBatchV1_2 gained majority on mainnet at 14:12 UTC. The reference node's rippled 3.4.0 did not recognise it; rippled 3.4.1, which does, is available from the XRPL Foundation's package repository (package xrpld 3.4.1), and the reference node was upgraded to it the same day (below). Unless it loses majority, the amendment activates about 2026-10-09 14:12 UTC; the engine must implement it once its source is published. BatchV1_1 lost majority for one voting round (ledgers 107,227,905 to 107,228,161, 14:29 to 14:46 UTC), which restarted its two-week period: Batch now activates about 2026-10-09 14:46 UTC, about half an hour after fixBatchV1_2, instead of 2026-09-29.
- **Second deployment (F402 and the vote change).** Build 0c76af1 replaced the live build at 10:53 EDT after its gate (1,640 suite tests and 707 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), again by warm restart: it rebuilt its state in 22.1 seconds and resumed one ledger behind the network. The health check matched 406 ledgers with no mismatches, and soak #31 began at 11:20 EDT. It recorded no receipts through the end of the day.
- **Reference node on rippled 3.4.1.** The reference node was upgraded from rippled 3.4.0 to 3.4.1 at 11:22 EDT and was fully synchronised again at 11:32, after re-acquiring the network's ledger; it recognises fixBatchV1_2. While it restarted, the validator switched to a public fallback server; a warm restart at 11:33 returned it to the reference node, 145 ledgers behind at resume and level with the network by 11:35, with no mismatches at any point. The engine's built-in copy of the rippled library stays at 3.4.0 until the 3.4.1 source is published; the two produce identical results until fixBatchV1_2 is enabled.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- fixBatchV1_2 gained majority — ledger 107,227,649, 2026-09-25 14:13 UTC
- BatchV1_1 lost majority — ledger 107,227,905, 2026-09-25 14:29 UTC
- BatchV1_1 gained majority — ledger 107,228,161, 2026-09-25 14:46 UTC

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Attested Reserve Invariant for Wrapped Bridge Assets** (interop) Add sfAttestedLockedBalance to the Bridge ledger object and a new witness-quorum transaction XChainAttestReserve, reusing the door account's existing
- **Atomic Multi-Leg Escrow Finish for DvP** (settlement) Add sfLinkedEscrows, a bounded (max 4) STVector256 of Escrow keylets, to EscrowFinish, and extend the transactor so that when present, doApply verifie
- **Credential-Gated Corporate Actions for MPT Issuers** (rwa) Add a CorporateAction ledger object (owner-reserved, keylet::corporateAction(issuanceID, sequence)) created by an MPTokenIssuance's issuer via a new M

Appendix — commits

Time	Repo	Commit	Subject
00:18	xrpl-validator	458c2c7	fix(node): validation relay answers the hubs' keep-alive pings
00:40	xrpl-validator	1c373c9	fix(node): voting-ledger validations keep their fields in canonical order
01:10	xrpl-validator	166f033	fix(ledger): F401 — AMM withdrawal reserve check counts XRP already paid out (soak #29 receipt)
10:07	xrpl-validator	e63a7d9	fix(ledger): F402 — TicketCreate paid with a Ticket does not count the ticket it spends
10:08	xrpl-validator	0c76af1	feat(node): stop voting for SingleAssetVault and LendingProtocol
10:31	xrpl-validator	677c18d	test(node): campaign 25 — Batch x Clawback, AMMClawback and token escrow (devnet vectors)